Example 9.2

Evaluate $\int_0^1 x\,dx$, as the limit of a sum.

Solution

Here $f(x)=x$, $a=0$ and $b=1$. Hence, we get

$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n} f\left(\frac{r}{n}\right) \Rightarrow \int_0^1 x\,dx = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\frac{r}{n}$$

$$= \lim_{n\to\infty}\frac{1}{n^2}\left[1+2+\cdots+n\right]$$

$$= \lim_{n\to\infty}\frac{1}{n^2}\frac{n(n+1)}{2} = \lim_{n\to\infty}\frac{1}{2}\left(1+\frac{1}{n}\right) = \frac{1}{2}.$$

Example 9.3

Evaluate $\int_0^1 x^3\,dx$, as the limit of a sum.

Solution

Here $f(x)=x^3$, $a=0$ and $b=1$. Hence, we get

$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n} f\left(\frac{r}{n}\right) \Rightarrow \int_0^1 x^3\,dx = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\frac{r^3}{n^3}$$

$$= \lim_{n\to\infty}\frac{1}{n^4}\left[1^3+2^3+\cdots+n^3\right] = \lim_{n\to\infty}\frac{1}{n^4}\frac{n^2(n+1)^2}{4}$$

$$= \lim_{n\to\infty}\frac{1}{4}\left(1+\frac{1}{n}\right)^2 = \frac{1}{4}.$$

Example 9.4

Evaluate $\int_1^4 \left(2x^2+3\right)dx$, as the limit of a sum.

Solution

We use the formula

$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\frac{b-a}{n}\sum_{r=1}^{n} f\left(a+(b-a)\frac{r}{n}\right)$$

Here $f(x)=2x^2+3$, $a=1$ and $b=4$.

So, we get

$$f\left(a+(b-a)\frac{r}{n}\right) = f\left(1+(4-1)\frac{r}{n}\right) = f\left(1+\frac{3r}{n}\right) = 2\left(1+\frac{3r}{n}\right)^2+3 = 5+\frac{18r^2}{n^2}+\frac{12r}{n}.$$

Hence, we get

$$\int_1^4 (2x^2+3)\,dx = \lim_{n\to\infty}\frac{3}{n}\sum_{r=1}^{n}\left(5+\frac{18r^2}{n^2}+\frac{12r}{n}\right) = \lim_{n\to\infty}\left[\frac{15}{n}\sum_{r=1}^{n}1+\frac{54}{n^3}\sum_{r=1}^{n}r^2+\frac{36}{n^2}\sum_{r=1}^{n}r\right]$$

$$= \lim_{n\to\infty}\left[\frac{15}{n}n+\frac{54}{n^3}\left(1^2+2^2+\cdots+n^2\right)+\frac{36}{n^2}\left(1+2+\cdots+n\right)\right]$$

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